


```
# Cell 1: Install Node.js in the Colab VM
!curl -fsSL https://deb.nodesource.com/setup_18.x | bash -
!apt-get install -y nodejs
```

2026-08-27 04:41:39 -

DEPRECATION WARNING

Node.js 18.x is no longer actively supported!
You will not receive security or critical stability updates for this version.

You should migrate to a supported version of Node.js as soon as possible.

Please see <https://nodesource.com/products/distributions> for details about which version may be appropriate for you.

The NodeSource Node.js distributions site contains information both about supported versions of Node.js and N|Solid supported Linux distributions. To learn more about usage, see:

<https://nodesource.com/products/distributions>

Continuing in 10 seconds ...

2026-08-27 04:41:49 - Installing pre-requisites

```
Hit:1 https://cli.github.com/packages stable InRelease
Get:2 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease [3
Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:4 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:5 https://r2u.stat.illinois.edu/ubuntu jammy InRelease [6,555 B]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Hit:7 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
Get:8 https://r2u.stat.illinois.edu/ubuntu jammy/main all Packages [10.8 MB]
Hit:9 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
Get:10 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packag
Get:11 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packag
Get:13 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Pack
Get:14 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4,
Get:16 https://r2u.stat.illinois.edu/ubuntu jammy/main amd64 Packages [3,177
Get:17 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages
Fetched 41.8 MB in 6s (7,586 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: Skipping acquire of configured file 'main/source/Sources' as repository 'h
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20260601~22.04.1).
gnupg is already the newest version (2.2.37-3ubuntu2.5).
```

```

gnupg is already the newest version (2.2.27-5ubuntu2.3).
gnupg set to manually installed.
Suggested packages:
  libcurl4-doc libidn11-dev libldap2-dev librtmp-dev
The following NEW packages will be installed:
  apt-transport-https
The following packages will be upgraded:
  curl libcurl4 libcurl4-openssl-dev
3 upgraded, 1 newly installed, 0 to remove and 22 not upgraded.

```

```

!curl -fsSL https://deb.nodesource.com/setup_18.x | bash -
!apt-get install -y nodejs

```

2026-08-27 04:43:44 -

DEPRECATION WARNING

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Continuing in 10 seconds ...

2026-08-27 04:43:54 - Installing pre-requisites

```

Hit:1 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease
Hit:2 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:3 https://cli.github.com/packages stable InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:6 https://r2u.stat.illinois.edu/ubuntu jammy InRelease
Hit:7 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:8 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:9 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
Hit:10 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
22 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: Skipping acquire of configured file 'main/source/Sources' as repository 'h
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20260601~22.04.1).
curl is already the newest version (7.81.0-1ubuntu1.27).

```

```
gnupg is already the newest version (2.2.27-3ubuntu2.5).
apt-transport-https is already the newest version (2.4.14).
0 upgraded, 0 newly installed, 0 to remove and 22 not upgraded.
Hit:1 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease
Hit:2 https://deb.nodesource.com/node\_18.x nodistro InRelease
Hit:3 https://cli.github.com/packages stable InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:6 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:7 https://r2u.stat.illinois.edu/ubuntu jammy InRelease
Hit:8 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:9 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
Hit:10 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
%%writefile server.js
const http = require('http');

const server = http.createServer((req, res) => {
  res.statusCode = 200;
  res.setHeader('Content-Type', 'text/plain');
  res.end('Hello World');
});

server.listen(3000, () => {
  console.log('Server chal raha hai port 3000 par');
});
```

Writing server.js

```
get_ipython().system_raw('node server.js &')
```

```
!curl http://localhost:3000
```

Hello World

```
!pkill -f "node server.js"
```

```
%%writefile server3.js
const http = require('http');
const url = require('url');

const PORT = 3000;

const server = http.createServer((req, res) => {

  // 1. Parse the URL to separate path and query parameters
  const parsedUrl = url.parse(req.url, true);
```

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const path = parsedUrl.pathname;
const query = parsedUrl.query;

console.log(`Request received: ${req.method} ${path}`);

// ----- ROUTE 1: Home page -----
if (path === '/' && req.method === 'GET') {
  res.statusCode = 200;
  res.setHeader('Content-Type', 'text/plain');
  res.end('Welcome! Try /greet?name=Rahul or /headers or POST to /data');
}

// ----- ROUTE 2: Query parameters -----
else if (path === '/greet' && req.method === 'GET') {
  const name = query.name || 'Guest';
  res.statusCode = 200;
  res.setHeader('Content-Type', 'text/plain');
  res.end(`Hello, ${name}! Welcome to the server.`);
}

// ----- ROUTE 3: Show request headers -----
else if (path === '/headers' && req.method === 'GET') {
  res.statusCode = 200;
  res.setHeader('Content-Type', 'application/json');
  res.end(JSON.stringify(req.headers, null, 2));
}

// ----- ROUTE 4: Handle POST data -----
else if (path === '/data' && req.method === 'POST') {
  let body = '';

  req.on('data', chunk => {
    body += chunk;
  });

  req.on('end', () => {
    res.statusCode = 201;
    res.setHeader('Content-Type', 'application/json');
    res.end(JSON.stringify({
      message: 'Data received successfully',
      yourData: body
    }));
  });
}

// ----- ROUTE 5: Custom status code demo -----
else if (path === '/error') {
  res.statusCode = 500;
  res.setHeader('Content-Type', 'text/plain');
  res.end('Simulated server error (500)');
```

```

    }

    // ----- DEFAULT: Route not found -----
    else {
      res.statusCode = 404;
      res.setHeader('Content-Type', 'text/plain');
      res.end('404 - Page Not Found');
    }
  });

  server.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
  });

```

Writing server3.js

```
get_ipython().system_raw('node server3.js &')
```

```

!echo "--- Home ---"
!curl http://localhost:3000/

!echo "\n--- Greet with name ---"
!curl "http://localhost:3000/greet?name=Rahul"

!echo "\n--- Headers ---"
!curl http://localhost:3000/headers

!echo "\n--- POST data ---"
!curl -X POST -d "Hello from client" http://localhost:3000/data

!echo "\n--- Error route ---"
!curl -i http://localhost:3000/error

!echo "\n--- Wrong route (404) ---"
!curl -i http://localhost:3000/wrongpath

```

```

--- Home ---
Welcome! Try /greet?name=Rahul or /headers or POST to /data\n--- Greet with n
Hello, Rahul! Welcome to the server.\n--- Headers ---
{
  "host": "localhost:3000",
  "user-agent": "curl/7.81.0",
  "accept": "/*/*"
}\n--- POST data ---
{"message":"Data received successfully","yourData":"Hello from client"}\n---
HTTP/1.1 500 Internal Server Error
Content-Type: text/plain
Date: Thu, 27 Aug 2026 04:53:35 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Content-Length: 28

```

```
Simulated server error (500)\n--- Wrong route (404) ---  
HTTP/1.1 404 Not Found  
Content-Type: text/plain  
Date: Thu, 27 Aug 2026 04:53:35 GMT  
Connection: keep-alive  
Keep-Alive: timeout=5  
Content-Length: 20
```

```
404 - Page Not Found
```